CSCE 3701 · TEAM CLOSET · MILESTONE 2 · SPRINT 1

Closet

Test Plan and Report · Version 2.0

Build v1.6 + Virtual Try-On (US-118) · November 4, 2026

1. Test plan

Objective. Show that every requirement in the SRS v2.0 works as specified, that nothing that worked before broke, and that the new Virtual Try-On is safe (another user cannot reach my photos) and fails gracefully.

Level	What	Tool	Where it runs
Unit	Pure rules: validators, recommendation engine, lifecycle state machine, Ask Closet parser, Virtual Try-On service and providers	Jest 29	every push (GitHub Actions)
API / integration	The real Express app through the security pipeline against a real PostgreSQL 16 database (emptied before each test)	Jest + Supertest	GitHub Actions (PostgreSQL service)
Acceptance	One test per Sprint 1 story, written from its acceptance criteria	Jest + Supertest	GitHub Actions
End-to-end (browser)	Real user journeys clicked in Chromium: demo, stateful, lifecycle, virtual try-on	Playwright (Python)	GitHub Actions + offline demo
Manual	Demo rehearsal on the presenting laptop; visual check of screens	—	team

Techniques. Equivalence partitioning and boundary values (photo size and type, 1 – 6 pieces, daily limit), state-transition testing (81 ordered pairs of the 9 lifecycle states; PENDING → PROCESSING → COMPLETED / FAILED), decision tables (deletion reasons), scenario tests (the black blazer at the tailor), and security cases (no token → 401, someone else's data → 404). **Test doubles:** the weather, Google sign-in and the AI try-on provider are replaced by fakes, so tests need no keys and no network. **Entry / exit:** a story is Done when its acceptance test and the full suite pass on GitHub Actions.

2. Results

Suite	Result	Run
Full suite, build v1.6 (unit + API + acceptance)	**599 / 599 passed** · 0 failed · line coverage 97.53 %	GitHub Actions CI #5, 29 Oct 2026 (real Jest, PostgreSQL 16)

Suite	Result	Run
Virtual Try-On unit tests VT-01 – VT-27	**23 / 23 passed**	sandbox, 4 Nov 2026 (Node 22; Jest-compatible runner because the npm registry is blocked there)
Full suite after Virtual Try-On (GitHub Actions, 4 Nov)	**630 / 631 passed** · 1 failed: OF-09 (defect D-33, fixed) · line coverage 94.53 %	GitHub Actions run 41865197065, real Jest + PostgreSQL 16
Virtual Try-On API tests API-VT01 – VT09	**9 / 9 passed**	GitHub Actions run 41865197065
e2e virtual_tryon_journey.py	virtual try-on journey passed on the offline demo: 13 steps	offline demo build, Chromium, 4 Nov 2026
e2e lifecycle_journey.py (regression)	offline demo: 11 checkpoints, 0 browser errors	offline demo build, 4 Nov 2026
e2e stateful_journey.py (regression)	offline demo: 14 checkpoints, 0 browser errors	offline demo build, 4 Nov 2026

Honest status. The old try-on tests TO-01 – TO-12 were removed with the old `/tryon` route they tested; the new Virtual Try-On replaces both. The real FASHN service has not been called yet (no API key in the sandbox): the provider was replaced by a test double and a mock. The live provider test is task T-30.

2.1 Virtual Try-On · unit tests (passed)

Test	What it shows
VT-01	FASHN: submits the person photo + garment with the category and returns the job id
VT-02	FASHN: status maps processing → PROCESSING, completed → COMPLETED + result URL, failed → FAILED
VT-03	FASHN: no key → disabled; bad key, rate limit and network errors become typed ProviderErrors
VT-04	provider choice comes from the environment; mock only when asked for; every provider has the full contract
VT-05	real type and pixel size are read from the file header; non-images are refused
VT-10	successful try-on: PENDING → PROCESSING → COMPLETED, result stored privately, original garment photo untouched
VT-11	an async provider is polled through GET: still PROCESSING until the job finishes
VT-12	invalid or someone else's wardrobe item → 400 / 404 (nothing is sent)
VT-13	unsupported garment type (shoes) → 422 UNSUPPORTED_GARMENT
VT-14	missing images: no user photo → PHOTO_REQUIRED; piece without a usable photo → GARMENT_IMAGE_MISSING / UNSUITABLE
VT-15	user photo validation: not an image → 415, too small → 422, no consent → 400

Test	What it shows
VT-16	provider failure → the try-on is FAILED with a friendly message, and Retry starts it again
VT-17	timeout, failed job and empty result all end as FAILED (never stuck, never blank)
VT-18	a temporary network error while polling keeps it PROCESSING (the next poll tries again)
VT-19	status retrieval and isolation: another user can neither read, save, retry nor delete my try-on
VT-20	save result, list saved, delete removes the stored images
VT-21	saved try-on photo: save, reuse, replace, delete
VT-22	uploading with "save this photo" also stores it for next time
VT-23	limits: daily quota and at most two running at once → 429
VT-24	provider not configured → 503 TRYON_DISABLED and status says enabled: false
VT-25	a saved look links the try-on to the outfit; someone else's look is 404
VT-26	an https result from the provider is copied into private storage
VT-27	account deletion removes every try-on file of the user

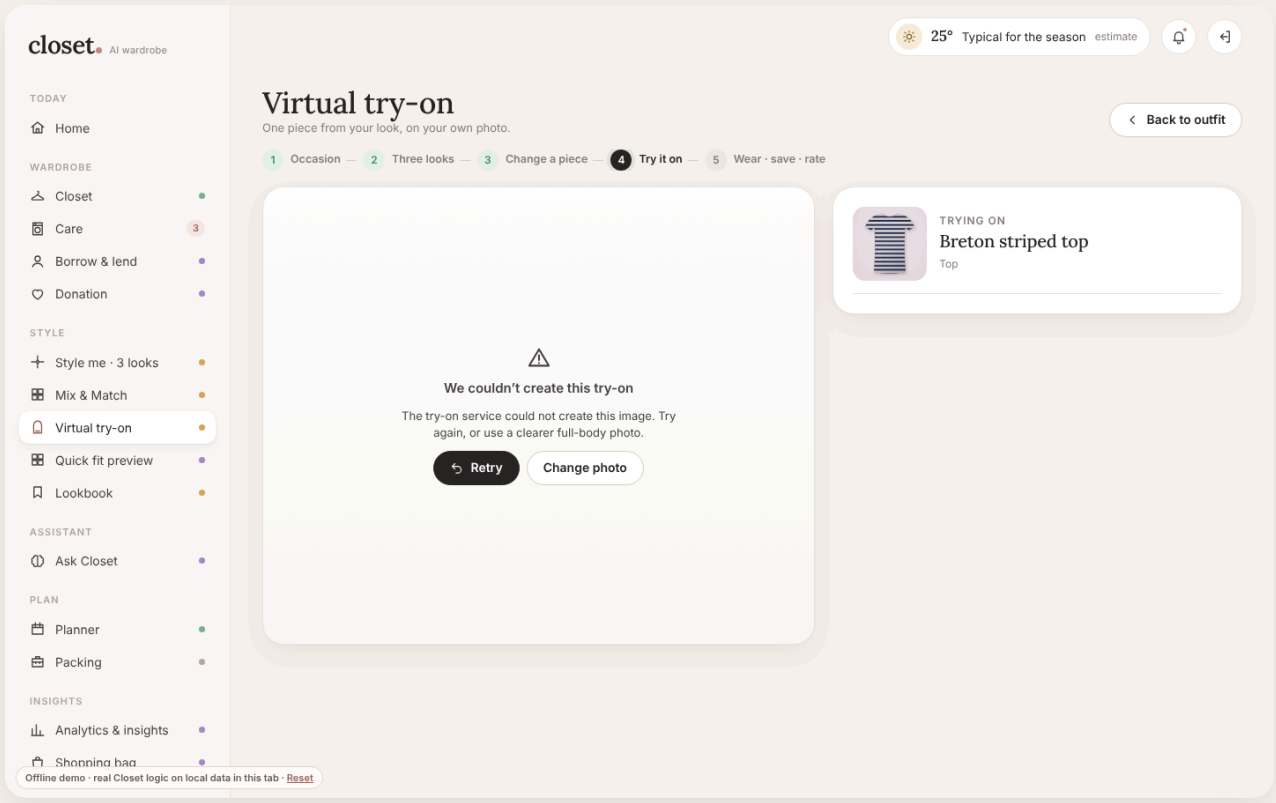
2.2 Virtual Try-On · API tests (run in CI)

Test	What it shows
API-VT01	without a login token every route is 401
API-VT02	create → 202 PROCESSING, then GET → COMPLETED with a signed result URL that serves an image
API-VT03	invalid item id → 400; unknown item → 404; shoes → 422 UNSUPPORTED_GARMENT
API-VT04	missing photo → 400 PHOTO_REQUIRED; a non-image → 415; no consent → 400
API-VT05	provider failure → FAILED with a friendly error, retry works once the provider is back
API-VT06	another user cannot read, save, retry or delete my try-on (404)
API-VT07	save the result, list saved, delete it
API-VT08	saved try-on photo: PUT, reuse with useSavedPhoto, DELETE
API-VT09	status route; with no provider configured, creating is 503

2.3 Virtual Try-On · browser journey

Step	Checked
v01	Every one of the 3 looks shows Virtual Try-On, Wear this, Save, Not for me and Preview
v02	The picker lists the look's pieces; only tops, jackets and dresses are enabled
v03 – v04	No photo → message; too-small photo and non-image rejected in the browser
v05	Generate without consent → message; consent + save photo
v06 – v07	Loading state, then the result with its label; Save result stored (API checked)
v08	Try another item offers the saved photo first

Step	Checked
v09 – v10	Forced provider failure → friendly error → Retry → success
v11	Leaving while it runs; “Your try-ons” lists it and it completes
v12	Back to outfit returns to the same three looks
v13	Delete saved photo (API checked)



Error state with Retry (step v09).

2.4 Defects found in this pass (all fixed)

ID	Found by	Severity	Description	Fix
D-29	e2e v05	High	Offline demo: deleting a seeded piece removed a photo shared by other pieces, so a top had “no usable photo”.	only photos added in the tab are removed
D-30	visual check	Medium	The garment picker opened inside the flipping outfit card and was clipped.	rendered into the page body (portal)
D-31	e2e v08	Low	Try another item on the same piece kept the previous upload instead of the saved photo.	reset photo choice and consent
D-32	e2e v12	Medium	Back to outfit from “Your try-ons” opened an empty Style me.	returns to the stored three looks
D-33	CI (OF-09)	Low	The seeded presentation plan had a fixed 24 °C; in November the seasonal estimate is 25 °C, so the “warmer forecast” alert no longer appeared.	plan = seasonal estimate − 6 °C, every month

2.5 Not verified yet

- Live FASHN generation (needs the team’s API key, T-30).
- The re-run of the full suite after the D-33 fix (expected 631 / 631).
- Usability test with classmates (T-33) and load test (T-34) — Sprint 2.